

# EduGenie – AI Learning Assistant

## Project Description:

EduGenie – AI Learning Assistant is an AI-powered educational platform that helps students learn concepts, ask questions, generate summaries, and receive detailed explanations using Generative AI. The application provides an easy-to-use interface where users can enter questions or topics and receive intelligent, context-aware responses.

Built using FastAPI, HTML, CSS, JavaScript, Python, and the Gemini API, EduGenie improves the learning experience by providing instant AI assistance, topic explanations, question answering, and text summarization through a simple web interface.

## Scenarios

### Scenario 1

A student enters a difficult programming question into EduGenie. The AI explains the concept with examples and simple language.

### Scenario 2

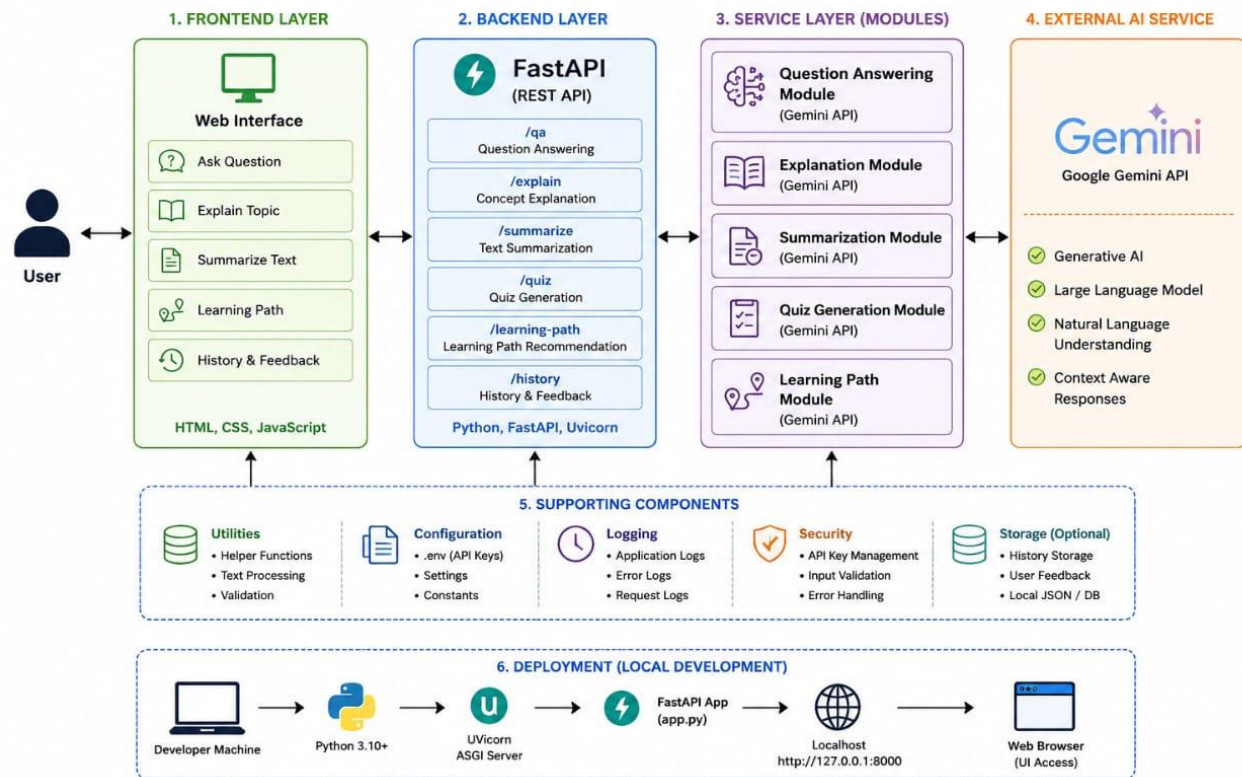
A learner uploads a long paragraph and requests a summary. EduGenie generates a concise summary highlighting the important points.

### Scenario 3

A student preparing for exams asks for an explanation of a topic. EduGenie provides a detailed explanation along with key concepts for revision.

## Technical Architecture:

### Technical Architecture: EduGenie – AI Learning Assistant



### Description:

The Frontend Layer provides a responsive web interface.

The Backend Layer processes user requests through FastAPI.

The AI Layer communicates with the Gemini API to generate intelligent responses.

The architecture is modular, scalable, and easy to maintain.

### Pre-requisites:

- Python Programming
- FastAPI Framework
- HTML
- CSS
- JavaScript
- REST API Basics

## Project Workflow:

### Milestone 1: Environment Setup

#### Activity 1.1: Define Requirements

Identify student learning requirements.

Define AI features.

Finalize project scope.

#### Activity 1.2: Design Architecture

Design frontend.

Design backend.

Define API flow.

#### Activity 1.3: Setup Environment

Install Python.

Install FastAPI.

Install required packages.

Configure Gemini API.

#### Activity 1.4: Create Project Structure

Create folders:

templates/

static/

services/

utils/

tests/

### Milestone 2: Backend Development

#### Activity 2.1: Develop FastAPI Backend

Create API endpoints:

/qa

/explain

/summarize

## Activity 2.2: Develop Question Answering Module

Responsibilities

Receive user questions

Generate AI answers

Return responses

## Activity 2.3: Develop Explanation Module

Responsibilities

Explain concepts

Provide detailed learning material

Generate examples

## Activity 2.4: Develop Summarization Module

Responsibilities

Summarize long text

Highlight key points

Produce concise output

## **Milestone 3: Frontend Development**

### Activity 3.1: Develop User Interface

Create pages for:

Home

Ask Question

Explain Topic

Summarize Text

Results Page

## Activity 3.2: Add User Features

Input validation

Loading animation

Copy response button

Responsive design

## **Milestone 4: AI Integration**

### Activity 4.1: Gemini API Integration

Responsibilities

Connect with Gemini API

Generate AI responses

Handle API requests

### Activity 4.2: Response Processing

Responsibilities

Format AI responses

Handle API errors

Improve response readability

## **Milestone 5: Testing**

### Activity 5.1: Unit Testing

Validate

Question Answering

Topic Explanation

Summarization

API Responses

### Activity 5.2: API Testing

Test

/qa endpoint

/explain endpoint

/summarize endpoint

## **Milestone 6: Deployment & Documentation**

### Activity 6.1: Local Deployment

Run

uvicorn app:app --reload

Open

http://127.0.0.1:8000

### Activity 6.2: Version Control

Repository contains

Source Code

Documentation

Images

Tests

README

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Application Pages

Home Page

Users can

Ask Questions

Explain Topics

Summarize Text

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Question Answering

Displays AI-generated answers for user questions.

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### Topic Explanation

Provides detailed explanations of concepts with examples.

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### Text Summarization

Generates concise summaries from lengthy text.

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### Results Page

Displays AI-generated output that can be copied for future use.

## **CONCLUSION:**

EduGenie – AI Learning Assistant demonstrates how Generative AI can improve digital learning by providing instant question answering, topic explanations, and text summarization. The application combines FastAPI, HTML, CSS, JavaScript, Python, and the Gemini API to deliver a responsive and intelligent learning platform.

Its modular architecture makes the system easy to maintain and extend. Future enhancements may include user authentication, learning history, multilingual support, PDF generation, cloud deployment, and voice-based interaction.